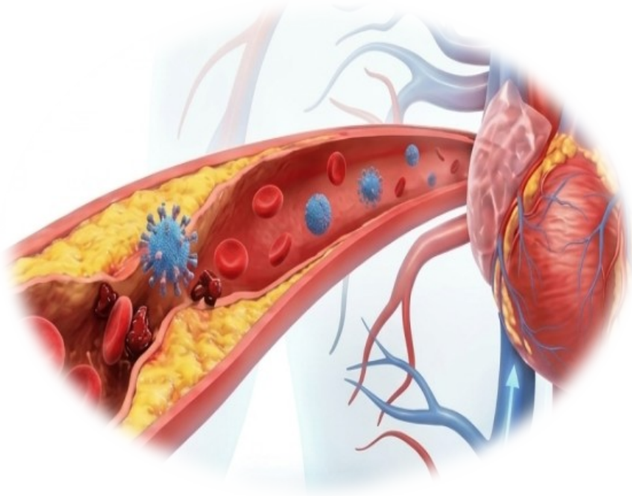
Hawler Medical University
College of Medicine

Determination of Serum (TG, T-Cholesterol), and Lipoproteins

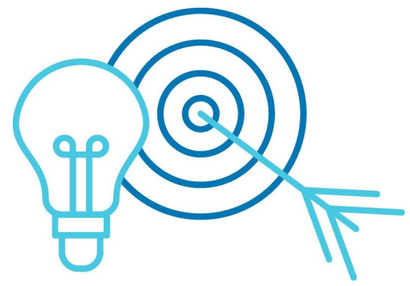
Biochemistry Lab 3
2nd-Stage
2026-2027



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Learning Objectives

- **Understand** the classification and medical importance of major blood lipids.
- **Explain** the analytical principles behind laboratory estimations of Triglycerides and Cholesterol.
- **Identify** the major lipoprotein types and differentiate between hyperlipidemia and hypolipidemia.
- **Apply** calculations like the Friedewald equation for LDL assessment.



Lipid Structure & Function

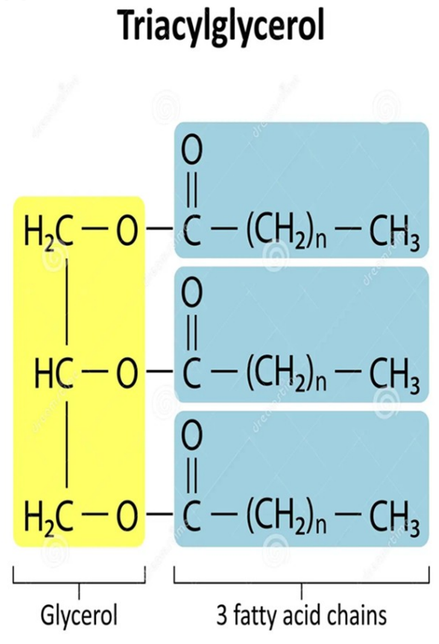
lipids are a heterogeneous class of hydrophobic organic molecules that are insoluble in water.

- Storage and Structural Lipids: Triacylglycerols serve as the primary long-term energy sources , while phospholipids and cholesterol esters form the structural foundation of cell membranes.
- Lipid processing breaks down fats into glycerol, free fatty acids, unesterified cholesterol, and ketone bodies, which serve as essential fuels and structural building blocks to drive the body's metabolic pathways.

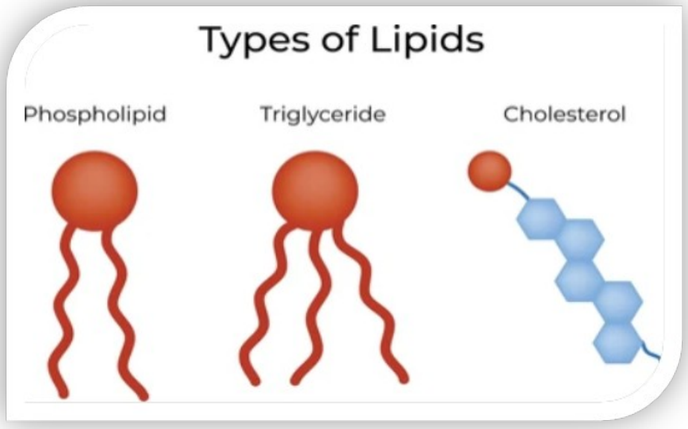
Profiles in Lipid Function: Triglycerides vs. Cholesterol

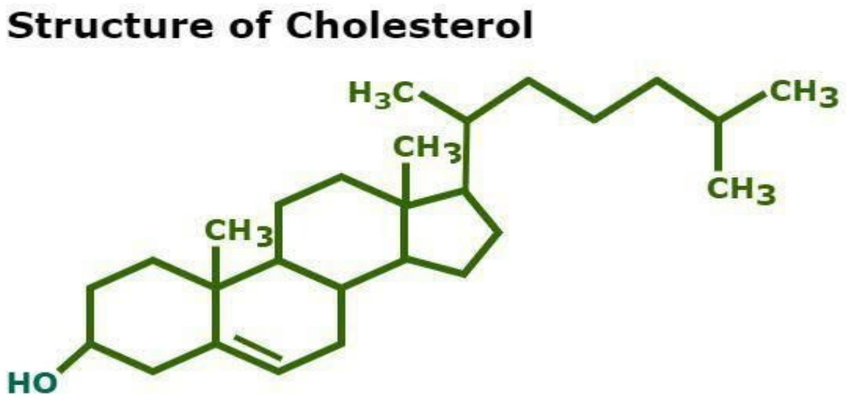
- Triglycerides function as the body's primary energy storage, while cholesterol serves as a critical structural component and metabolic precursor.

- **Triglycerides:** Glycerol backbones esterified to three fatty acids that form the main constituent of dietary and stored fats, serving as vital metabolic energy reservoirs.



- **Cholesterol:** A structural sterol synthesized by major hubs (liver, brain, adrenal cortex) that modulates cell membrane fluidity and acts as the essential baseline for synthesizing steroid hormones, bile acids, and vitamins. It circulates in blood plasma primarily as cholesteryl esters.





Lipoprotein Structure and Function

- **Lipoproteins** overcome lipid insolubility by encapsulating hydrophobic lipids within an amphipathic shell, allowing them to travel easily through the bloodstream.
- These molecules consist of a hydrophobic core surrounded by a hydrophilic shell of phospholipids, cholesterol, and proteins.
- Surface proteins called apolipoproteins (Apo A, B, C, E, D) maintain the particle's structure and direct how lipids are metabolized in the body.

Major Lipoprotein Classes

Lipoprotein classification relies on an inverse relationship between size and density:

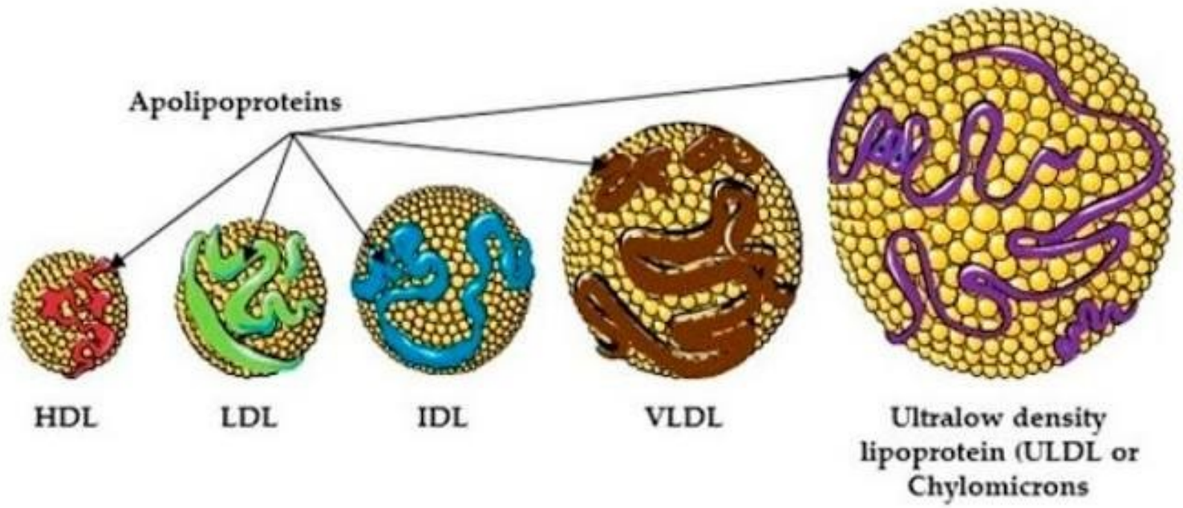
a) Chylomicrons

b) Very low density lipoproteins (VLDL)

c) Intermediate density lipoproteins (IDL)

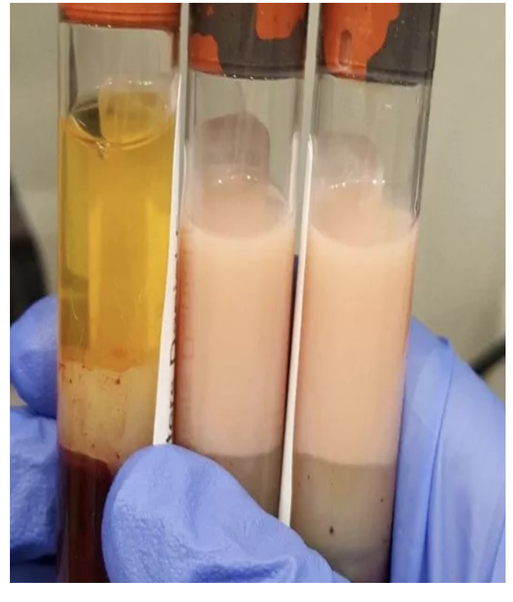
d) Low density lipoproteins (LDL)

e) High density lipoproteins (HDL).



Sample Collection & Preparation

- ✓ **Sample Type:** Fresh serum or plasma (collected in EDTA or heparin tubes).
- ✓ **Fasting Requirement:** 9 to 12-hour overnight fast is mandatory.



Principle:

A. Triglycerides (TG): measured at 500-520 (nm).

$$TG \xrightarrow{\text{Lipase}} \text{Glycerol + FFAs}$$

$$\text{Glycerol + ATP} \xrightarrow{\text{Gk}} \text{Glycerol-3-p + ADP}$$

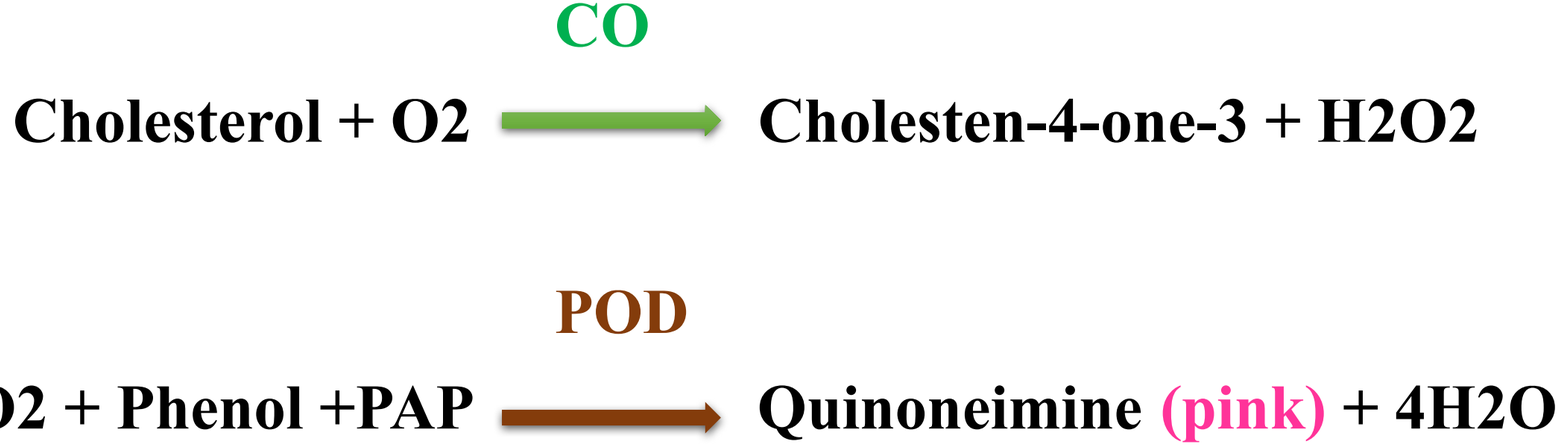
$$\text{Glycerol-3-p + O2} \xrightarrow{\text{GPO}} \text{Dihydroxy acetone-P + H2O2}$$

$$\text{H2O2 + 4-Chlorophenol + PAP} \xrightarrow{\text{POD}} \text{Quinoneimine + H2O}$$

Principle:

B. Total Cholesterol (TC): measure light absorbance at (500–550) nm.





The Friedewald Formula: LDL & VLDL Estimation

Friedewald Equation: is a mathematically estimates LDL and VLDL cholesterol from a standard lipid panel, avoiding costly direct testing.

- VLDL-C Estimation = Triglycerides (in mg/dL) ÷ 5

- LDL-C Estimation: Total Cholesterol - HDL- VLDL

Reference Values of Lipid Profile

Lipid Component	Normal Range
Total Cholesterol	Less than 200 mg/dL
LDL Cholesterol (Bad)	Less than 100 mg/dL
HDL Cholesterol (Good)	Men: Above 40 mg/dL Women: Above 50 mg/dL
Triglycerides	Less than 150 mg/dL
VLDL Cholesterol	5 to 40 mg/dL

Clinical Classification of Hyperlipidemia

Hyperlipidemia: Excess circulating levels of total cholesterol, LDL cholesterol, or triglycerides (TG) in the bloodstream.

1. Hypertriglyceridemia (Elevated TGs)

- **Primary (Familial):** Caused by genetic mutations like lipoprotein lipase (LPL) deficiency, leading to decreased catabolism of VLDL and chylomicrons.

- **Secondary:** Driven by obesity, Diabetes Mellitus (DM), kidney diseases (nephrotic syndrome, renal failure), hypothyroidism, and excessive alcohol intake.

Clinical Classification of Hyperlipidemia

2) Hypercholesterolemia (Elevated Cholesterol)

- **Primary (Familial):** Genetic mutations causing LDL receptor deficiency, preventing cells from clearing LDL from the blood.

- **Secondary:** High saturated fat/cholesterol diet, obesity, alcohol intake, and underlying diseases (Atherosclerosis, DM, Nephrotic Syndrome, Obstructive Jaundice, Hypothyroidism).

Clinical Classification of Hyperlipidemia

- **Clinical Implications**

  Hyperlipidemia can cause two main problems

  • Cardiovascular risk: High lipids can cause atherosclerosis, heart disease, and stroke.

  • Pancreatic risk: Very high triglycerides can cause acute pancreatitis.

  Management

  Management focuses on lowering lipid levels

  • Reduce saturated fat in the diet.

  • Exercise regularly.

  • Statins are mainly used to lower cholesterol.

  • Fibrates are mainly used to lower high triglycerides

Hyperlipidemia (high cholesterol)

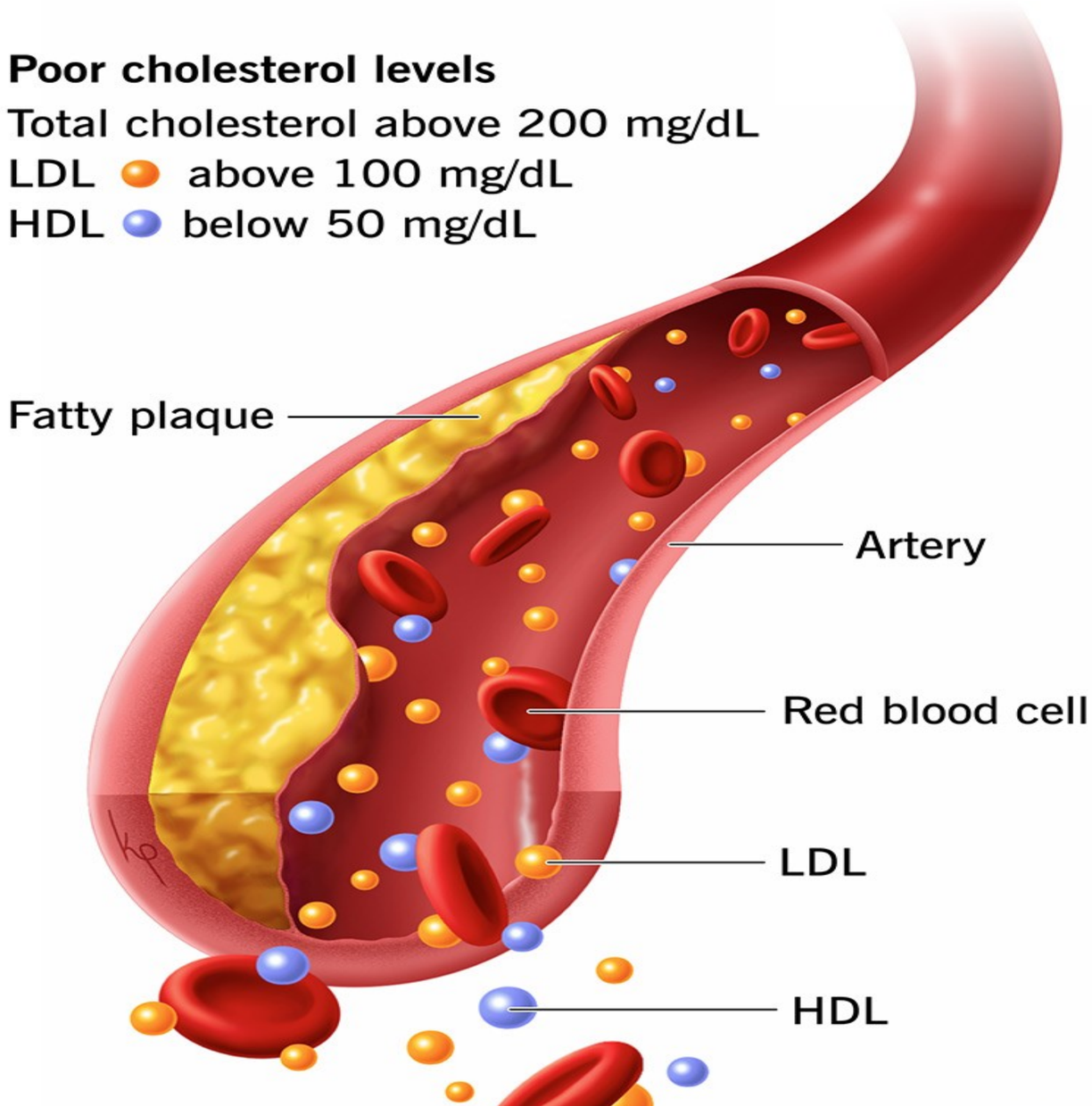
Excess of lipids or fat in the blood

Poor cholesterol levels

Total cholesterol above 200 mg/dL

LDL above 100 mg/dL

HDL below 50 mg/dL



Clinical Classification of Hypolipidemia

Hypolipidemia: is a condition where circulating blood lipids are abnormally low, typically defined as having a total cholesterol level below 120–130 mg/dL.

1) Primary (Genetic): Rare inherited conditions like Abetalipoproteinemia and Hypobetalipoproteinemia cause a deficiency in apolipoprotein B (apo B). This deficiency results in a total absence of chylomicrons, VLDL, and LDL, leading to exceptionally low plasma triglycerides and cholesterol. Another genetic cause is Tangier Disease.

Clinical Classification of Hypolipidemia

2) **Secondary:** Acquired conditions including malnutrition, starvation, severe malabsorption, hyperthyroidism, and chronic or severe liver diseases (like severe hepatitis).

- **Clinical Implications:** The lack of lipid transport vehicles can cause fat-soluble vitamin deficiencies (A, D, E, and K), neurological symptoms (such as ataxia and neuropathy), and acanthocytosis (abnormally shaped red blood cells).

Lab Procedure

- Incubation: **Mix 10 µL serum + 1 mL reagent** and **leave for 5 mins at 37°C** (or 10 mins at room temperature) until it turns pinkish-red.
- Measurement: Zero the spectrophotometer with a reagent blank, then read the standard and patient sample at 500–520 nm.

$$\text{Concentration of Sample} = \frac{\text{Absorbance of Sample}}{\text{Absorbance of Standard}} \times \text{Concentration of st}$$

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Nauck, M., Warnick, G. R., & Rifai, N. (2002). Methods for measurement of LDL-cholesterol: a critical assessment of direct assays. *Clinical Chemistry*, 48(2), 236–254.

Rare Disease & Genetic Databases

- **MedlinePlus Genetics.** *Abetalipoproteinemia*. National Library of Medicine. Updated August 2020. medlineplus.gov
- **National Organization for Rare Disorders (NORD).** *Abetalipoproteinemia*. NORD Rare Disease Database. Published 2020. rarediseases.org
- **Welty, F. K. (2018).** Abetalipoproteinemia. In: Adam, M. P., Feldman, J., Mirzaa, G. M., et al., editors. *GeneReviews*. University of Washington, Seattle. Updated October 25,
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Online Academic Curriculums

- **NCBI Bookshelf.** *Clinical Methods: The History, Physical, and Laboratory Examinations* — Chapter 31: Cholesterol, Triglycerides, and Associated Lipoproteins.
- **Medicine LibreTexts.** *Section 2.7: Lipids - Triglycerides, Phospholipids and Sterols.*

Thank you for your time. May our shared knowledge inspire future innovation

Thank you!